WORKED SOLUTION

STEP II 1989, Question 4

Pure • Rational Function Sketching

STEP II 1989, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q4

The key idea

A product or quotient of linear factors is easiest to differentiate logarithmically, because the log turns it into a sum. For the range, ask the question backwards: for which z does $\mathrm{f}(x) = z$ have a real solution? That equation is a quadratic in x, so the answer comes from a discriminant. The sketches then follow from the zeros, the poles, the behaviour for large $|x|$, and whether or not some values of z are missed.

One word on the printed text. The condition $a + d \neq c + d$ only says $a \neq c$, which we already know. The condition actually used is $a + b \neq c + d$, and we assume that throughout.

Solution

The derivative

Where $\mathrm{f}(x) \neq 0$, that is for $x \neq a, b$ (as well as $x \neq c, d$), we can take logarithms of moduli:

$$\ln|\mathrm{f}(x)| = \ln|x-a| + \ln|x-b| - \ln|x-c| - \ln|x-d|.$$

Differentiating, and using $\frac{\mathrm{d}}{\mathrm{d}x}\ln|x-k| = \frac{1}{x-k}$,

$$\frac{\mathrm{f}'(x)}{\mathrm{f}(x)} = \frac{1}{x-a} + \frac{1}{x-b} - \frac{1}{x-c} - \frac{1}{x-d}.$$

Now multiply by x. For $x \neq 0$ each term becomes

$$\frac{x}{x-k} = \frac{1}{1-k/x} = \left(1 - \frac{k}{x}\right)^{-1},$$

which gives the required identity

$$\frac{x\,\mathrm{f}'(x)}{\mathrm{f}(x)} = \left(1-\frac{a}{x}\right)^{-1} + \left(1-\frac{b}{x}\right)^{-1} - \left(1-\frac{c}{x}\right)^{-1} - \left(1-\frac{d}{x}\right)^{-1}.$$

Large $|x|$

When $|x|$ is much larger than $|k|$, the binomial expansion gives $(1-k/x)^{-1} = 1 + \frac{k}{x} + O\left(\frac{1}{x^2}\right)$. The four leading 1s cancel ($1+1-1-1=0$), leaving

$$\frac{x\,\mathrm{f}'(x)}{\mathrm{f}(x)} = \frac{a+b-c-d}{x} + O\left(\frac{1}{x^2}\right), \qquad \text{so} \qquad \mathrm{f}'(x) = \frac{\mathrm{f}(x)}{x^2}\left[(a+b-c-d) + O\left(\tfrac{1}{x}\right)\right].$$

Since $a+b-c-d \neq 0$, the bracket has the sign of $a+b-c-d$ once $|x|$ is large enough. Also $\mathrm{f}(x) \to 1$ as $|x| \to \infty$, so $\mathrm{f}(x) > 0$ there, and $x^2 > 0$. Hence $\mathrm{f}'(x)$ has the same sign as $a+b-c-d$ for $|x|$ large, at both ends of the real line.

It's also worth recording which side of the asymptote $y = 1$ the curve is on. Directly,

$$\mathrm{f}(x) - 1 = \frac{(c+d-a-b)x + ab - cd}{(x-c)(x-d)} \approx \frac{c+d-a-b}{x} \quad \text{for large } |x|,$$

and the numerator shows that $\mathrm{f}(x) = 1$ at exactly one point, $x_0 = \dfrac{cd-ab}{c+d-a-b}$.

The range condition: a method

Rearrange $\mathrm{f}(x) = z$ to $(x-a)(x-b) = z(x-c)(x-d)$, which is the quadratic

$$(1-z)x^2 - (a+b-(c+d)z)x + (ab - cdz) = 0.$$

- No root of this can be $x = c$ or $x = d$, because the left-hand side $(x-a)(x-b)$ is non-zero there. So real roots of the quadratic are exactly the real solutions of $\mathrm{f}(x) = z$.
- For $z \neq 1$ there is a real root if and only if the discriminant $\Delta(z) = \big(a+b-(c+d)z\big)^2 - 4(1-z)(ab-cdz)$ is ≥ 0. Multiply out, multiply by $(c-d)^2 > 0$, and collect powers of z. With $P = (a-c)(b-d)$ and $Q = (a-d)(b-c)$, the useful identity is $P - Q = (a-b)(c-d)$, so the constant term $(a-b)^2(c-d)^2$ equals $(P+Q)^2 - 4PQ$. The given inequality is then exactly $(c-d)^2\Delta(z) \geq 0$.
- For $z = 1$ the equation is linear with x-coefficient $c+d-a-b \neq 0$, so there is a root; and the inequality holds since both sides differ by $(c-d)^2(a+b-c-d)^2 > 0$.

The sketches

Write $G(z) = \big[(c-d)^2 z + P + Q\big]^2 - 4PQ$. As z varies, the square takes every value in $[0, \infty)$, so the smallest value of G is $-4PQ$. Therefore:

$$\text{the range of f is all of } \mathbb{R} \iff PQ \leq 0.$$

If $PQ > 0$, the values of z with $G(z) < 0$ form an open interval, and these are the missed values.

Ask yourself How do we know the sign of PQ without the actual numbers?

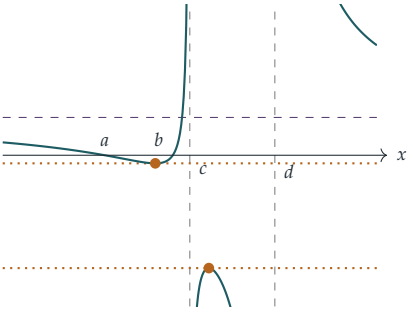
Regroup: $PQ = (a-c)(a-d)\cdot(b-c)(b-d)$. Since $a < c < d$, the first pair is a product of two negatives, so it's positive. So PQ has the sign of $(b-c)(b-d)$, which is negative exactly when b lies between the poles c and d. Everything depends on where b sits relative to $c < d$, and on the sign of $a+b-c-d$:

1. $a < b < c < d$: then $a+b < c+d$ automatically.
2. $a < c < b < d$: again $a+b < c+d$, since $a < c$ and $b < d$.
3. $a < c < d < b$ with $a+b < c+d$.
4. $a < c < d < b$ with $a+b > c+d$.

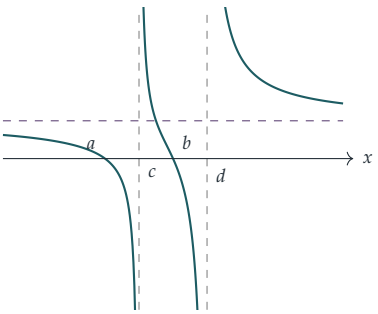
So the range is all of $\mathbb{R}$ only in case 2. In cases 1, 3 and 4 an interval of values is missed.

One more tool helps with the turning points. If f has a stationary point at x_1 with $f(x_1) = z_1$, then x_1 is a repeated root of $(x-a)(x-b) - z_1(x-c)(x-d) = 0$. (For $z_1 = 1$ this is linear, so it can't happen.) A repeated root means $G(z_1) = 0$. So the stationary values are exactly the ends of the missing interval. In case 2, $G > 0$ always, so there are no stationary points at all.

The sign of f just either side of each pole comes from counting negative factors. In each picture the vertical dashed lines are the poles $x = c$, $x = d$, the horizontal dashed line is the asymptote $y = 1$, and the dotted amber lines mark the ends of the missing interval, which are the values at the stationary points (amber dots). The curves are drawn for particular values (given under each picture), but the shapes are the same for every choice in that case.



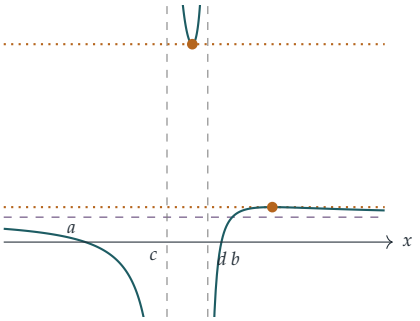
Case 1: $a, b, c, d = -2, 0, \frac{1}{2}, 3$



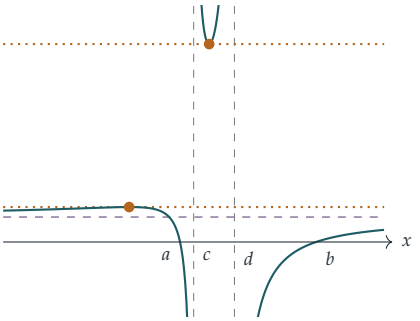
Case 2: $a, b, c, d = -1, 1, 0, 2$

Case 1 ($a < b < c < d$). Here f < 1 far to the left and f > 1 far to the right. On $x < c$ the curve comes in just below 1, falls to 0 at a, dips to a negative minimum between a and b, and rises through b to $+\infty$ at c. Between the poles both denominator factors have opposite signs and the numerator is positive, so f $\to -\infty$ at both ends, with a negative local maximum in between. To the right of d the curve falls from $+\infty$ to 1. The local maximum on (c, d) lies below the local minimum on (a, b), and the values strictly between them are missed. *Range not all of* $\mathbb{R}$.

Case 2 ($a < c < b < d$). There are no stationary points. The curve falls from just below 1 through a to $-\infty$ at c; then from $+\infty$ through b to $-\infty$ at d; then from $+\infty$ down to 1. The middle branch alone takes every real value. *Range is all of* $\mathbb{R}$.



Case 3: $a, b, c, d = -3, 2, 0, \frac{3}{2}$



Case 4: $a, b, c, d = -\frac{1}{2}, \frac{9}{2}, 0, \frac{3}{2}$

Case 3 ($a < c < d < b$, $a + b < c + d$). On $x < c$ the curve falls from just below 1 through a to $-\infty$. Between the poles the numerator is negative, so $f \to +\infty$ at both ends, with a local minimum m. To the right of d it rises from $-\infty$ through b, crosses $y = 1$ at x_0, reaches a local maximum M just above 1, and then falls back to 1 from above. The values in (M, m) are missed, where $1 < M < m$. *Range not all of* $\mathbb{R}$.

Case 4 ($a < c < d < b$, $a + b > c + d$). This is case 3 with the outer behaviour reversed. Far to the left f is just above 1 and increasing. It reaches a local maximum $M > 1$, crosses $y = 1$ at $x_0 < a$, falls through a to $-\infty$ at c. The middle branch is a U with minimum m, as before. To the right of d the curve rises from $-\infty$ through b towards 1 from below. Again (M, m) is missed. *Range not all of* $\mathbb{R}$.

Checks

- The identity: a finite-difference estimate of $xf'(x)/f(x)$ at random a, b, c, d, x agrees with the right-hand side to about 10^{-9}.
- Case 1 with $a, b, c, d = -2, 0, \frac{1}{2}, 3$: here $P = \frac{15}{2}$, $Q = \frac{5}{2}$, $(c-d)^2 = \frac{25}{4}$, and $G(z) = 0$ gives $z = -\frac{8}{5} \pm \frac{4}{5}\sqrt{3}$, about -0.2144 and -2.9856. These are exactly the stationary values found by solving $f'(x) = 0$ (at $x \approx -0.515$ and $x \approx 1.060$). Sampling f at 4×10^5 points never lands strictly between them.
- Cases 3 and 4 (the drawn values): the missing interval is $(1.4007, 7.9327)$, again matching the stationary values. In case 2, $f'(x) = 0$ has no real solutions.

Common slips

- Forgetting that $f(x) = z$ is only a quadratic when $z \neq 1$: the value $z = 1$ needs its own sentence.

- Assuming the curve approaches $y = 1$ from the same side at both ends. It doesn't: $\mathrm{f} - 1 \approx (c + d - a - b)/x$ changes sign with x.

Beyond STEP

Turning "what is the range of f?" into "for which z does $\mathrm{f}(x) = z$ have a solution?" is the idea of a preimage, and the observation that stationary values are where two solutions merge is the start of the theory of critical values. A rational function of degree 2 like this one is a two-to-one map, and its behaviour is far cleaner over the complex numbers, where every value is taken (counting multiplicity) exactly twice. Needham's *Visual Complex Analysis* develops this geometric viewpoint beautifully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.